

PREDICTING MIE COEFFICIENTS FOR WELL-MIXED CLOUD PARTICLES USING MACHINE LEARNING

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Take Home Message

- Transmission spectra from JWST observations show evidence of mixed cloud particles in exoplanet clouds.
- Our neural network accurately reproduces mie coefficients and key optical features five times faster on average than normal calculations.

OBSERVATIONS

With JWST, it is now possible to observe a diversity of cloud materials within exoplanet atmospheres. For example, the effect of silicate clouds is observed in transmission spectra [1]. Modeling these atmospheres is integral to understanding these planets and interpreting their observations; however, modeling complex mixtures in exoplanet clouds is **computationally intensive** and time-consuming.

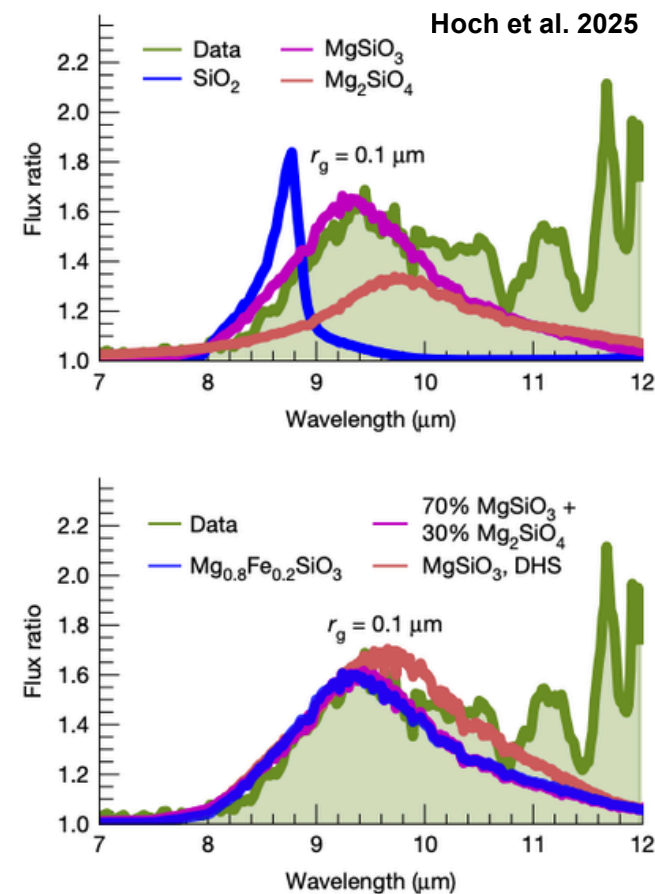
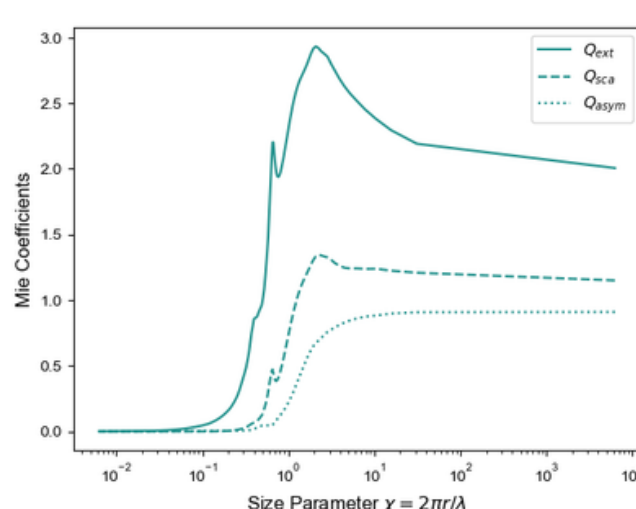
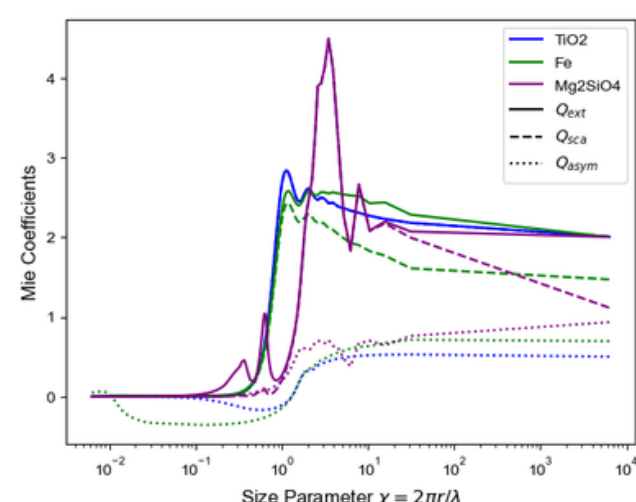
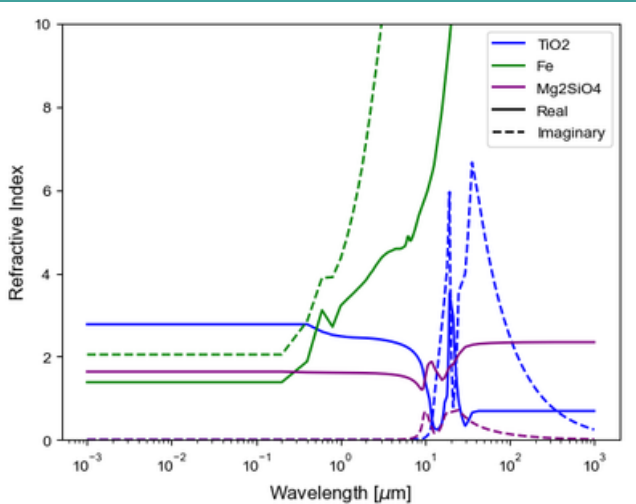


Figure 1: Flux ratio of the observed spectra of YSES-1 c, a Jupiter-like planet orbiting a young, solar mass star, to a self-consistent atmospheric model. Hoch et al. (2025) find that including silicate species and mixtures resolves this difference.

SCATTERING AND MIXING THEORY

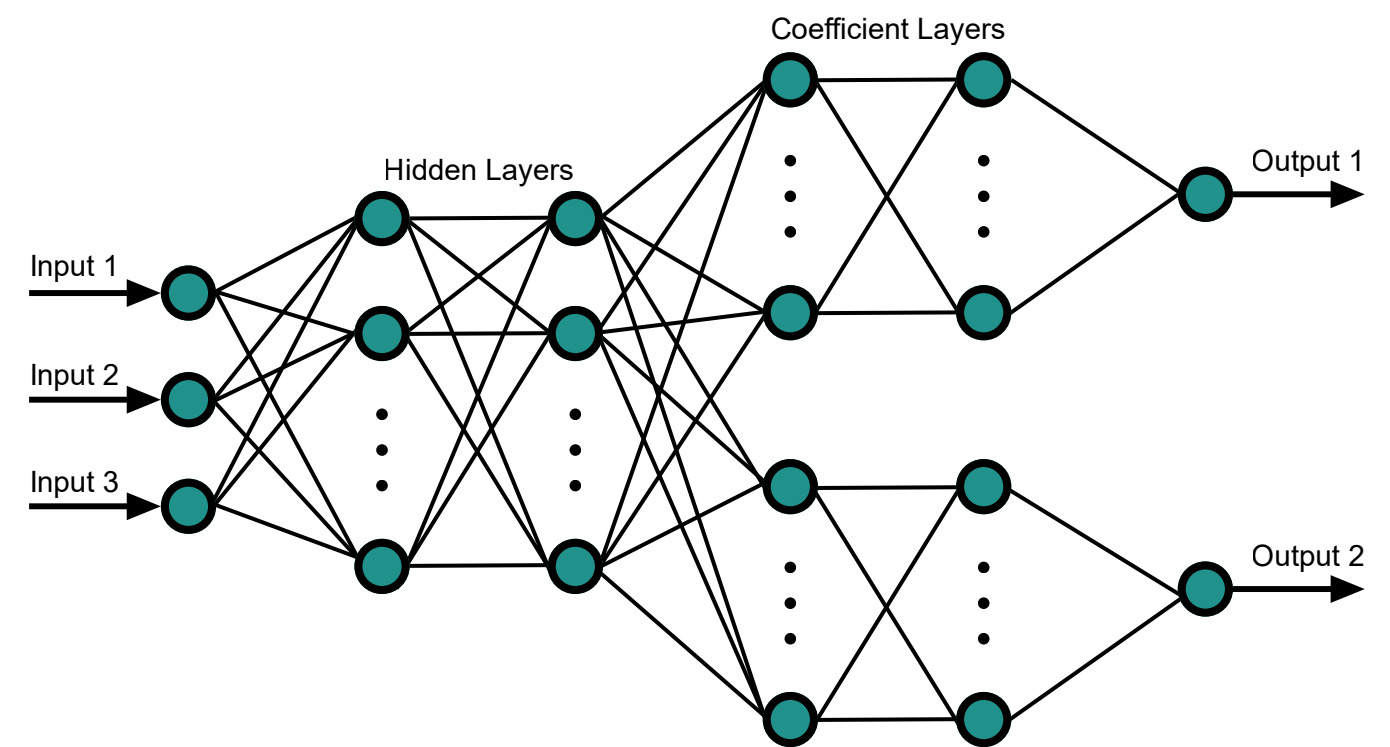


Mie theory [2] describes how light interacts with spherical particles of a similar size to the wavelength of light using coefficients, such as extinction, scattering, and asymmetry. These coefficients are dependent on the radius of the scatter, the wavelength of the incoming light, and the refractive index of the scatter. Using Mie theory in combination with the LLL Approximation [3], which calculates the effective refractive index for a mixture of elements and compounds assuming well-mixed particles, we can calculate the opacity of exoplanet clouds with diverse compositions.

Figure 2: a) Real and imaginary refractive indices for each species as a function of wavelength. b) Extinction, scattering, and asymmetry coefficients for each species versus size parameter, with cloud particle radius of one micron. c) Mie coefficients for a forsterite-dominated (90%) mixture with a cloud particle radius of one micron.

NEURAL NETWORK

Neural networks use neurons to process information, learn patterns in the data, and make predictions. This model is highly customizable and can handle **complex, non-linear data**, making it the ideal choice for opacity calculations. [4]



To test the viability of using machine learning for opacity calculations, we optimize a neural network for a cloud mixture of titanium oxide, iron, and forsterite. This mixture is physically realistic, computationally challenging, and should result in a defined feature in the model spectrum at a wavelength of 10 microns from forsterite.

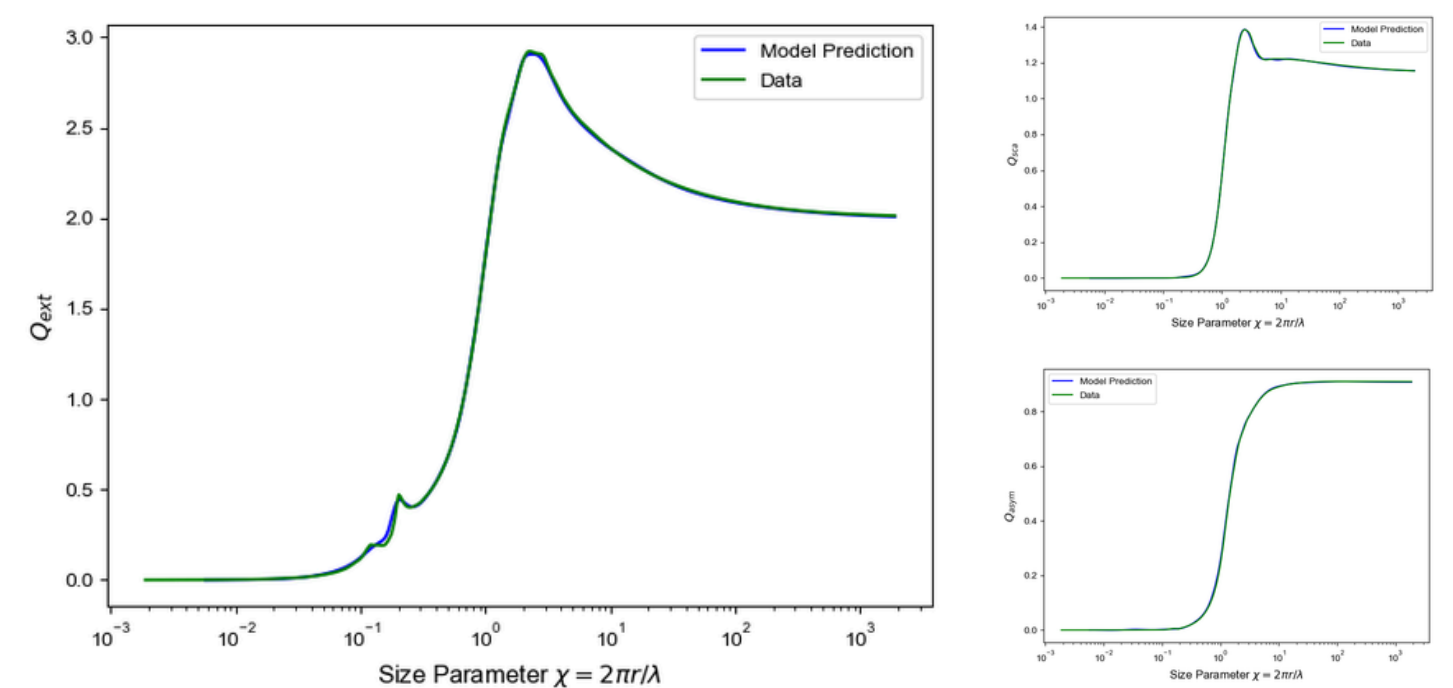


Figure 2: a) Extinction coefficient for a forsterite-dominated (90%) mixture with cloud particle radius of 0.3 microns from normal calculations, in green, and as predicted by the model, in blue. b) Scattering coefficient for the same mixture and cloud particle radius. c) Asymmetry coefficient for the same mixture and cloud particle radius.

We find that the network successfully fits the coefficient curves and reproduces the 10 micron feature, with computational time approximately **five times faster** on average than normal calculations.

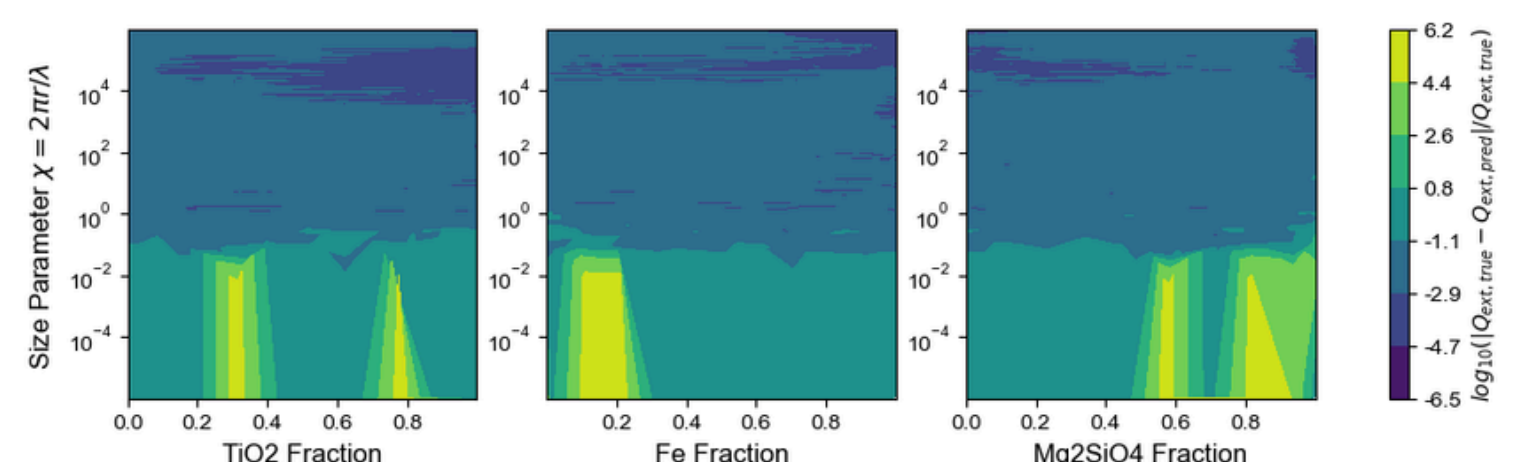


Figure 4: Relative residuals, in logscale, of the extinction coefficient versus size parameter and each material fraction.

We will continue to optimize this model to improve accuracy and prediction speed, and test the model predictions by reproducing transmission spectra. Neural networks will be created for various species mixtures and made publicly available for cloud modeling.

References:

- [1] Hoch, K.K.W., Rowland, M., Petrus, S. et al. Silicate clouds and a circumplanetary disk in the YSES-1 exoplanet system. *Nature* 643, 938–942 (2025).
- [2] Mie, G. (1908), Beiträge zur Optik trüber Medien, speziell kolloidaler Metallösungen. *Ann. Phys.*, 330: 377-445.
- [3] Landau, L. D., & Lifshitz, E. M. (1960). *Electrodynamics of continuous media*.
- [4] LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. *Nature* 521, 436–444 (2015).